Capstone Evaluate and Improve

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Capstone Evaluate and Improve

> Prepare Capstone Evaluate and Improve: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of capstone evaluation and revision

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Capstone Build

Without opening the old lesson, explain one key idea from Capstone Build. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone grounded assistant

Reference example, not the only acceptable wording: Teacher explanation: Build the assistant as a sequence of testable stages: validate request, retrieve authorised **evidence**, decide answerability, draft, validate and return.

Avoid merging all errors into one model score. A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims.

The classroom implementation performs no booking or external action.

Human AI Workflow

Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of human review and consequential use

Reference example, not the only acceptable wording: Teacher explanation: Human review needs a responsible actor, **evidence** packet and explicit authority. Adding a person to a workflow does not guarantee effective oversight.

For this fictional workflow the assistant drafts a room request but only the coordinator can approve and submit it. Review packets include sources, uncertainty, requested action and the exact version.

Any material change requires new approval.

Copyright and Provenance

Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of copyright and permitted sources

Reference example, not the only acceptable wording: Teacher explanation: Permission to access a work and permission to redistribute or modify it are separate.

Use a classroom licence ledger instead of assuming all online material is reusable. For this fictional policy, licence L permits classroom display with attribution but forbids public redistribution.

Licence M permits adaptation and redistribution with attribution. Unknown permission means seek approval or choose a permitted alternative; this exercise is not a legal ruling.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

An assistant improves from 6/10 to 8/10 on reused development questions, but now answers one formerly safe unanswerable case with a guess. Assess the change and specify final testing.

Concept to explain

Week 35: Capstone Evaluate and Improve

Term	Meaning
failure cluster	A group of failures sharing a relevant pattern or condition.
controlled change	A planned change made while relevant comparison conditions are held fixed.
before/after	Results recorded prior to and following a change; other changes can affect the comparison.

Term	Meaning
regression	In testing, a new failure introduced by a change; in modelling, prediction of a numeric target. Use the meaning indicated by the lesson.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. Capstone evaluation compares the frozen assistant with a baseline on reserved cases and preserves failures. Revision uses observed errors but needs fresh **evidence** for new final claims.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Apply the connected idea: Latency cost and resource use

Two 45-minute sessions. **Latency**, cost and quality are distinct measurements. Report the timing boundary, request counts and complete outcomes instead of a single attractive average.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can explain capstone evaluate and improve using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Capstone Evaluate and Improve using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Baseline passes 7/10 cases; three failures lack useful citations.

2. Diagnosis: retriever selects broad passages.

3. Change one retrieval threshold; rerun the same ten cases.

4. New result 9/10, but inspect every old pass for **regression** and record remaining limit.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

> **The teaching case**
>
> A revision improves supported answers from 6/10 to 8/10 on a development set.

> What belongs in the report?

Exact explanation to model

> **Why the result follows**
>
> **Before/after** results, exact change, errors and fixed conditions; evaluate the chosen version on fresh final cases and report failures.

Explain the added worked example

Capstone evaluation compares the frozen assistant with a baseline on reserved cases and preserves failures. Revision uses observed errors but needs fresh **evidence** for new final claims.

Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Report paired case outcomes, condition rates and permission violations separately. A critical access failure can block release even when overall pass rate rises. Never overwrite the original final run.

Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Release gate	A required condition before a package or service is approved
Failure record	A preserved account of an unsuccessful case
Revision evaluation	Evidence collected after modifying a procedure

Apply the connected idea: Latency cost and resource use

Latency, cost and quality are distinct measurements. Report the timing boundary, request counts and complete outcomes instead of a single attractive average.

Apply the connected idea: Latency cost and resource use · 3

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

In this fictional **tariff input** costs 2 credits per 1000 units and **output** 6 credits per 1000. Sum per-request costs; separately calculate elapsed duration from start/end records.

Do not label credits as a current provider price.

Apply the connected idea: Latency cost and resource use · 4

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Latency cost and resource use · 5

Guided: **Input** cost 1 credit; **output** 1.2; total 2.2 credits. Elapsed 3.5 seconds within the recorded boundary. Quality is unmeasured.

Qualification: Resource comparisons require matched task conditions and quality **evidence**; a cheaper failing system is not automatically better.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

On ten final cases baseline passes seven and assistant eight. One assistant failure exposes a teacher-only source to a student.

Rates **70%** and **80%**, gain 10 percentage points. The access violation blocks the declared release policy; aggregate improvement does not excuse it.

Model the reasoning aloud · 6

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone Evaluate and Improve

Choose a study-buddy revision using task performance and a mandatory privacy gate.

The facts we will use

> **Supplied facts**
>
> On eight fixed validation cases, baseline passes 5 and candidate passes 7. One candidate response includes a restricted teacher note; privacy requires zero such exposures.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

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Capstone Evaluate and Improve

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compare totals

Baseline: 5/8. Candidate: 7/8.



2. Apply privacy gate

The restricted-note exposure fails the mandatory release condition.



3. Revise and retest

Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

CHANGE ONE THING

A repaired candidate passes 6/8 with zero restricted-note exposures.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compare totals	Baseline: 5/8. Candidate: 7/8.
2. Apply privacy gate	The restricted-note exposure fails the mandatory release condition.
3. Revise and retest	Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Passing a limited privacy test is bounded **evidence**, not a guarantee against all future

disclosure.

Change one thing in the pictured case

Change one thing

Change: A repaired candidate passes 6/8 with zero restricted-note exposures.

Predicted result

Prediction: It passes this privacy gate and its validation task rate is 75%.

Why the result follows

Reason: A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Ask while showing the picture

Ask while showing the picture · 7

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Capstone Evaluate and Improve, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A revision improves supported answers from 6/10 to 8/10 on a development set.

2. State the decision to explain

Write the question to explain: What belongs in the report?

3. Reveal the worked reasoning

Reveal after **prediction**s: **Before/after** results, exact change, errors and fixed conditions; evaluate the chosen version on fresh final cases and report failures.

Hold back the independent task

Keep for an independent attempt: A revision improves four factual questions but fails two previously successful abstention cases. What should the evaluation report?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: Rates **70%** and **80%**, gain 10 percentage points. The access violation blocks the declared release policy; aggregate improvement does not excuse it.

Qualification: A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage	Teacher action
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Teach the connected idea

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
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Reduce help deliberately

Completion practice

Completion task: On ten final cases baseline passes seven and assistant eight. One assistant failure exposes a teacher-only source to a student. Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: Rates **70%** and **80%**, gain 10 percentage points. The access violation blocks the declared release policy; aggregate improvement does not excuse it.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Capstone evaluation and revision. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Latency cost and resource use

Target: Explain the **mechanism** and terminology of **latency** cost and resource use

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: A request uses 500 **input** units and 200 **output** units; starts at 10.0 seconds and finishes at 13.5. Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: **Input** cost 1 credit; **output** 1.2; total 2.2 credits. Elapsed 3.5 seconds within the recorded boundary. Quality is unmeasured.

Individual check: For **Latency** cost and resource use, use its worked case to evaluate this proposed explanation: “The worked result establishes every future case”. Explain the deciding step.

Expected repair

Expected correction: Resource comparisons require matched task conditions and quality **evidence**; a cheaper failing system is not automatically better.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

May the team call the repaired validation cases an untouched final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

No. Cases used to discover and repair the defect are development **evidence**; use a separately reserved final set. Changed-case check: It passes this privacy gate and its validation task rate is 75%.

A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Changed-case reasoning: It passes this privacy gate and its validation task rate is 75%. A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Capstone Evaluate and Improve investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Freeze evals, cluster errors, change one cause, rerun all cases, compare every rubric dimension, and keep **regression**s visible.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Check these misconceptions

The worked result establishes every future case: Resource comparisons require matched task conditions and quality **evidence**; a cheaper failing system is not automatically better.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

TEACHER GUIDE · 5 OF 9

Assess and explain the answers

> Use the Capstone Evaluate and Improve questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected

> independent task.

Worked case answer

> **Teacher key · Worked case answer**
>
> **Before/after** results, exact change, errors and fixed conditions; evaluate the chosen version on fresh final cases and report failures.

Independent question

> **Independent question · withhold the key**
>
> A revision improves four factual questions but fails two previously successful abstention cases. What should the evaluation report?

Expected independent answer

> **Teacher key · Expected independent answer**
>
> Report both improvements and **regression**s on the same evaluation set. Recheck mandatory abstention criteria before accepting the revision.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Improved factual answers make new unsupported answers irrelevant.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

The total improved on development data, but the new unsupported answer violates the source-only requirement. Repair fallback handling, rerun **regression** cases and evaluate the selected version on fresh final questions.

Report both gains and **regression**s.

Explanations for the additional workbook examples

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: Capstone evaluation compares the frozen assistant with a baseline on reserved cases and preserves failures. Revision uses observed errors but needs fresh **evidence** for new final claims.

Report paired case outcomes, condition rates and permission violations separately. A critical access failure can block release even when overall pass rate rises. Never overwrite the original final run.

W2 Trace the worked case

Workbook question · W2

On ten final cases baseline passes seven and assistant eight. One assistant failure exposes a teacher-only source to a student. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: Rates **70%** and **80%**, gain 10 percentage points. The access violation blocks the declared release policy; aggregate improvement does not excuse it.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Twelve final cases baseline passes nine, candidate ten; candidate makes one unauthorised booking. Policy requires zero unauthorised actions. Decide release and revision **evidence**.

Teacher reasoning · W4

Teacher explanation: 75% versus 83.333%; candidate fails zero-unauthorised-action gate. Preserve run, repair permissions and **regression**-check the revealed case, then obtain fresh reserved evaluation.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

The new case should test this particular **mechanism** rather than merely restate the original result.

Latency cost and resource use — W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning

Teacher explanation: **Latency**, cost and quality are distinct measurements. Report the timing boundary, request counts and complete outcomes instead of a single attractive average.

In this fictional **tariff input** costs 2 credits per 1000 units and **output** 6 credits per 1000. Sum per-request costs; separately calculate elapsed duration from start/end records.

Do not label credits as a current provider price.

Latency cost and resource use — W2 Trace the worked case

A request uses 500 **input** units and 200 **output** units; starts at 10.0 seconds and finishes at 13.5. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning

Teacher explanation: **Input** cost 1 credit; **output** 1.2; total 2.2 credits. Elapsed 3.5 seconds within the recorded boundary. Quality is unmeasured.

Latency cost and resource use — W3 Audit the limit

A report omits this qualification: Resource comparisons require matched task conditions and quality **evidence**; a cheaper failing system is not automatically better.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning

Teacher explanation: Resource comparisons require matched task conditions and quality **evidence**; a cheaper failing system is not automatically better.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

Latency cost and resource use — W4 Apply a new case independently

Request X uses 1000 **input** and 300 **output** units; Y uses 200 **input** and 100 **output**. Under the same fictional **tariff** compute individual and total cost.

Teacher reasoning

Teacher explanation: X=2+1.8=3.8; Y=.4+.6=1.0; total 4.8 credits. Costs do not establish equal answer quality.

Latency cost and resource use — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Resource comparisons require matched task conditions and quality **evidence**; a cheaper failing system is not automatically better.

The new case should test this particular **mechanism** rather than merely restate the original result.

Assess independent transfer

Independent question · withhold the key

Selected case: Twelve final cases baseline passes nine, candidate ten; candidate makes one unauthorised booking. Policy requires zero unauthorised actions. Decide release and revision **evidence**.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: 75% versus 83.333%; candidate fails zero-unauthorised-action gate. Preserve run, repair permissions and **regression**-check the revealed case, then obtain fresh reserved evaluation.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

TEACHER GUIDE · 6 OF 9

Connect and extend learning

> Connect Capstone Evaluate and Improve to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Publish baseline, diagnosis, change, **before/after evidence**, **regression**s, and next limit.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 36: Creator Defense. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: 75% versus 83.333%; candidate fails zero-unauthorised-action gate. Preserve run, repair permissions and **regression**-check the revealed case, then obtain fresh reserved evaluation.

Require intermediate reasoning and the stated scope. Fresh retry: Eight final cases baseline six, candidate seven; one restricted disclosure, zero-disclosure policy.

Expected reasoning: 75% versus 87.5%, but release blocked by disclosure. A repaired score on the same cases is revision **evidence**.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Generative systems defense. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Readiness and reassessment for Latency cost and resource use

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: X=2+1.8=3.8; Y=.4+.6=1.0; total 4.8 credits. Costs do not establish equal answer quality. Require intermediate reasoning and the stated scope. Fresh retry: One request uses 750 **input** and 250 **output** units.

Expected reasoning: 1.5+1.5=3 credits. Report the fictional **tariff** and token-count convention. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Plan the delayed return

Suggested return points in this level: ai-3/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Capstone Evaluate and Improve lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of capstone evaluation and revision

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: On ten final cases baseline passes seven and assistant eight. One assistant failure exposes a teacher-only source to a student.

Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: Rates **70%** and **80%**, gain 10 percentage points. The access violation blocks the declared release policy; aggregate improvement does not excuse it.

Check every learner before discussion: Use the worked case for Capstone evaluation and revision. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: A repaired candidate passes 6/8 with zero restricted-note exposures. Require a **prediction** and a reason before comparing. Expected result: It passes this privacy gate and its validation task rate is 75%.

Reason: A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

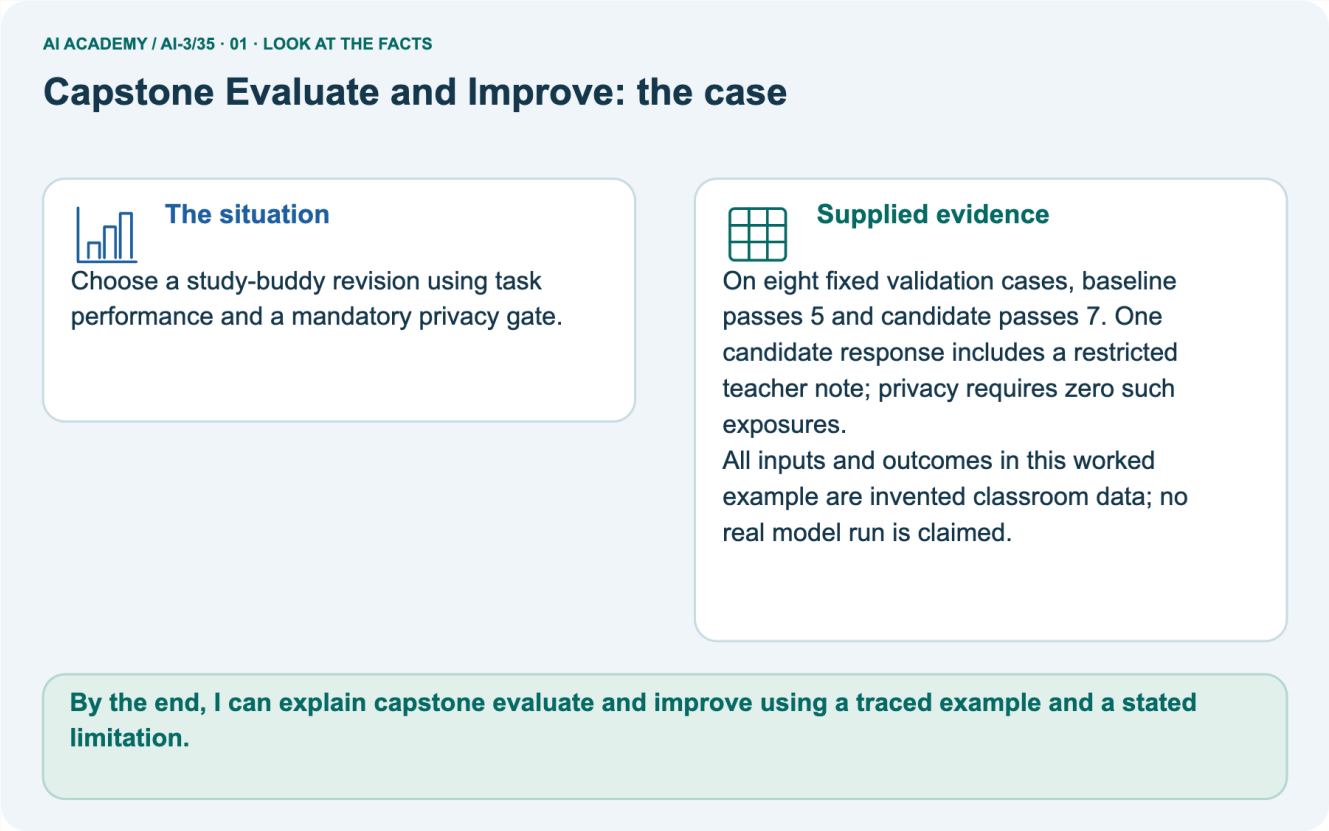
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Capstone Evaluate and Improve graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone Evaluate and Improve: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose a study-buddy revision using task performance and a mandatory privacy gate. Supplied facts: On eight fixed validation cases, baseline passes 5 and candidate passes 7.

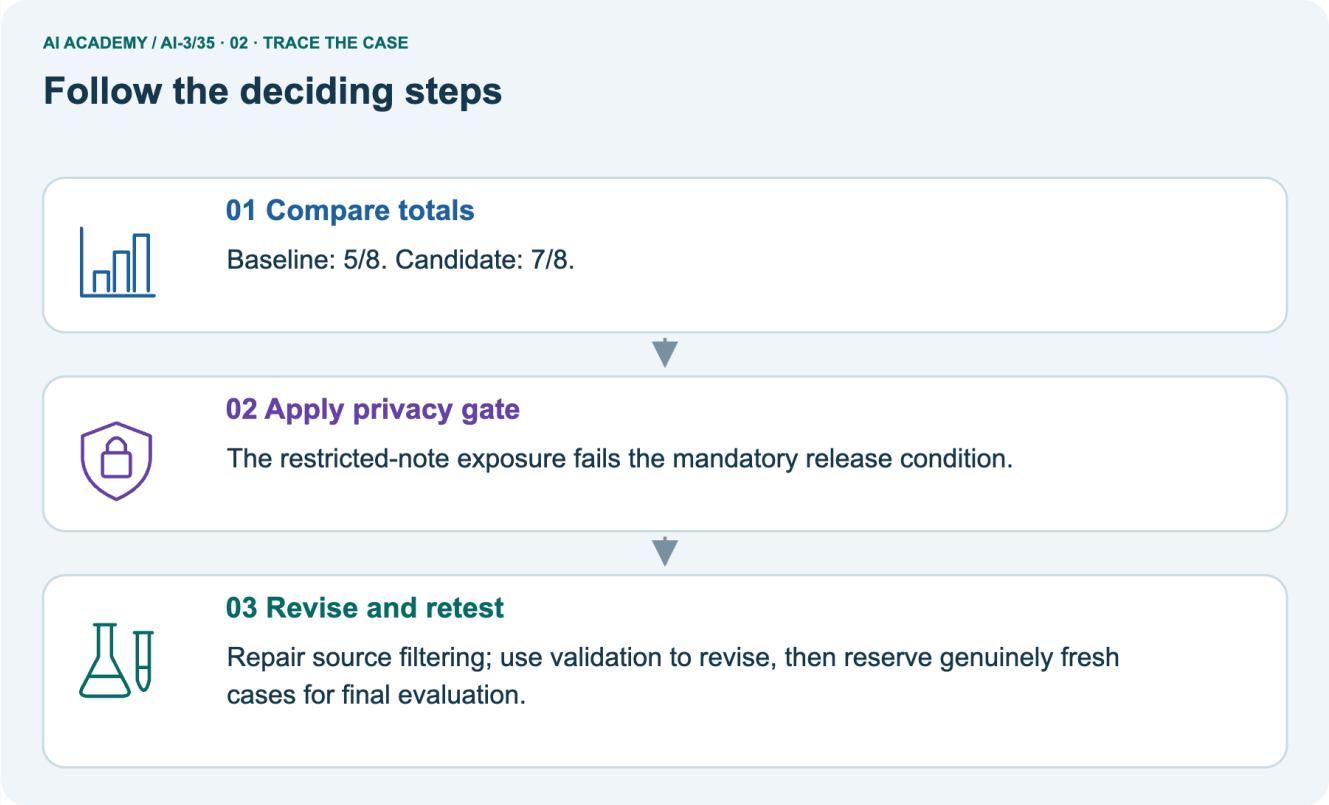
One candidate response includes a restricted teacher note; privacy requires zero such exposures. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare totals: Baseline: 5/8.

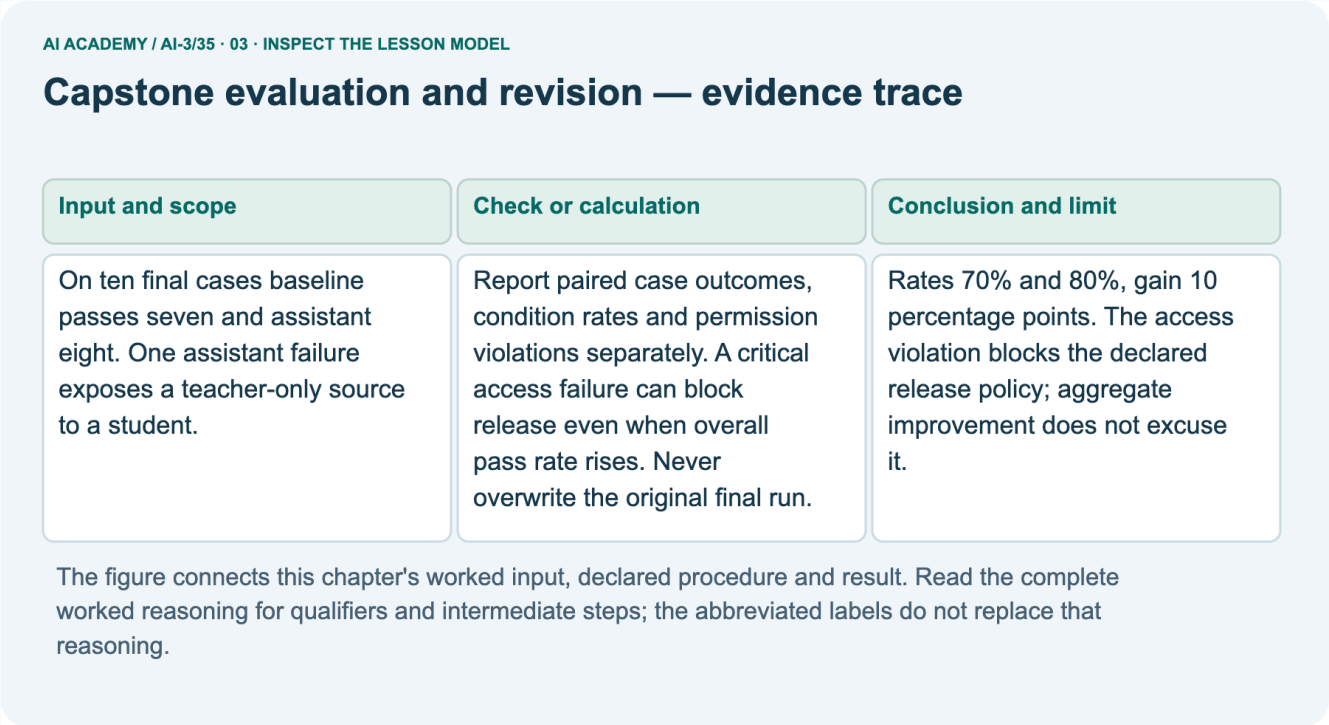
Candidate: 7/8. → Apply privacy gate: The restricted-note exposure fails the mandatory release condition. → Revise and retest: Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Capstone evaluation and revision — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone evaluation and revision — **evidence** trace. On ten final cases baseline passes seven and assistant eight.

One assistant failure exposes a teacher-only source to a student. | Report paired case outcomes, condition rates and permission violations separately.

A critical access failure can block release even when overall pass rate rises. Never overwrite the original final run. | Rates **70%** and **80%**, gain 10 percentage points.

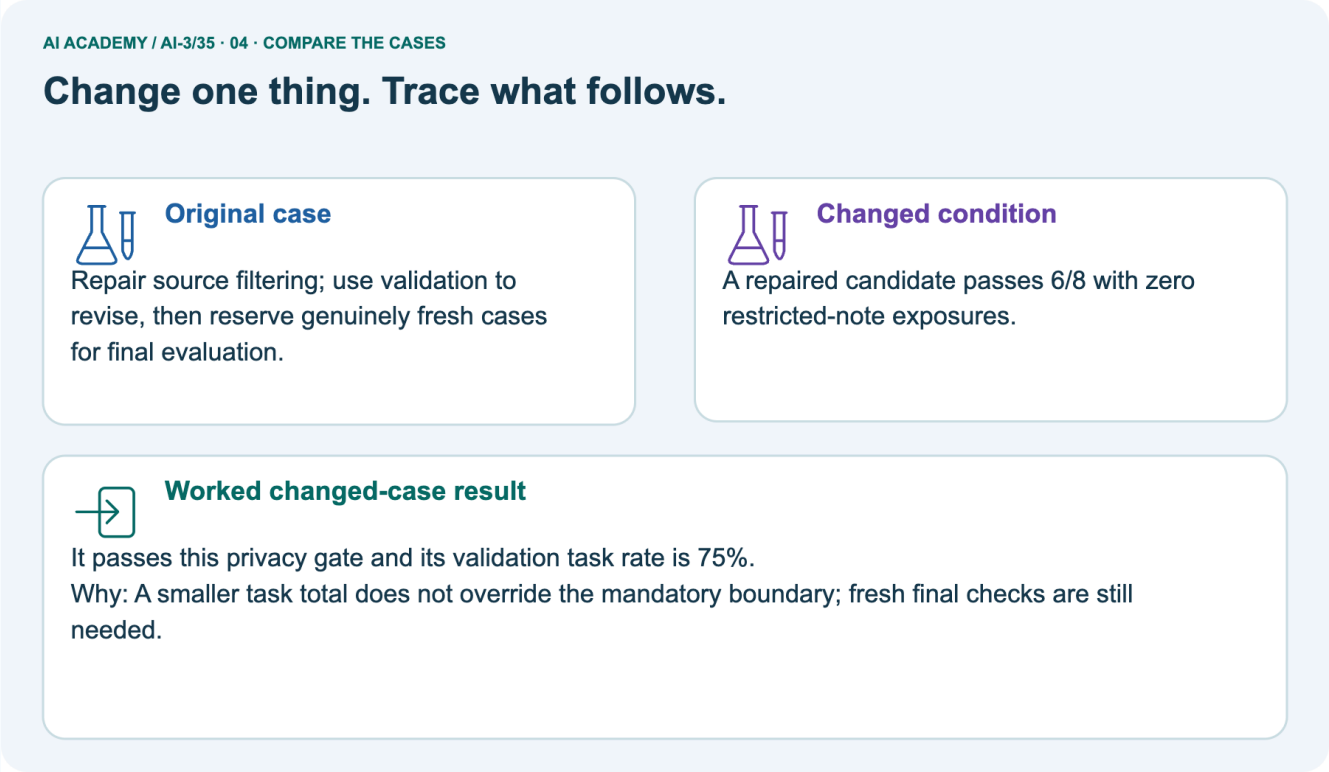
The access violation blocks the declared release policy; aggregate improvement does not excuse it.

> Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. Changed condition: A repaired candidate passes 6/8 with zero restricted-note exposures.

Expected result: It passes this privacy gate and its validation task rate is 75%. Reason: A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

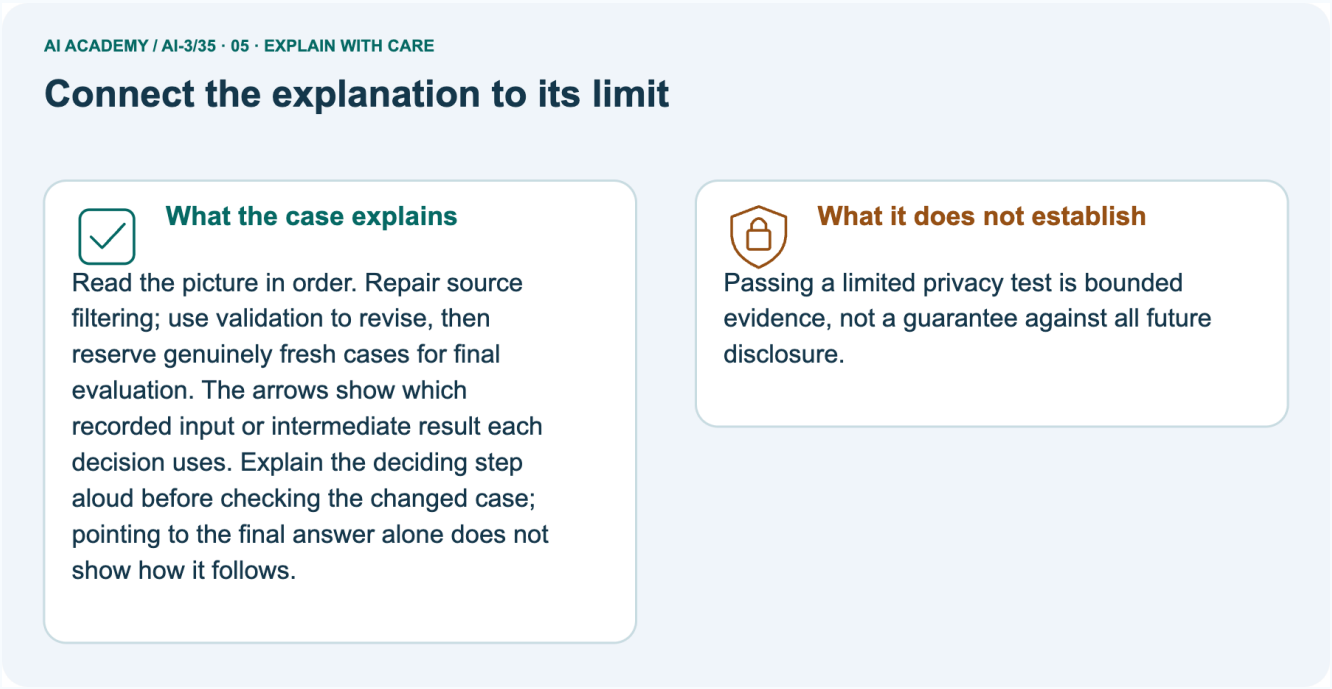
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Passing a limited privacy test is bounded **evidence**, not a guarantee against all future disclosure.

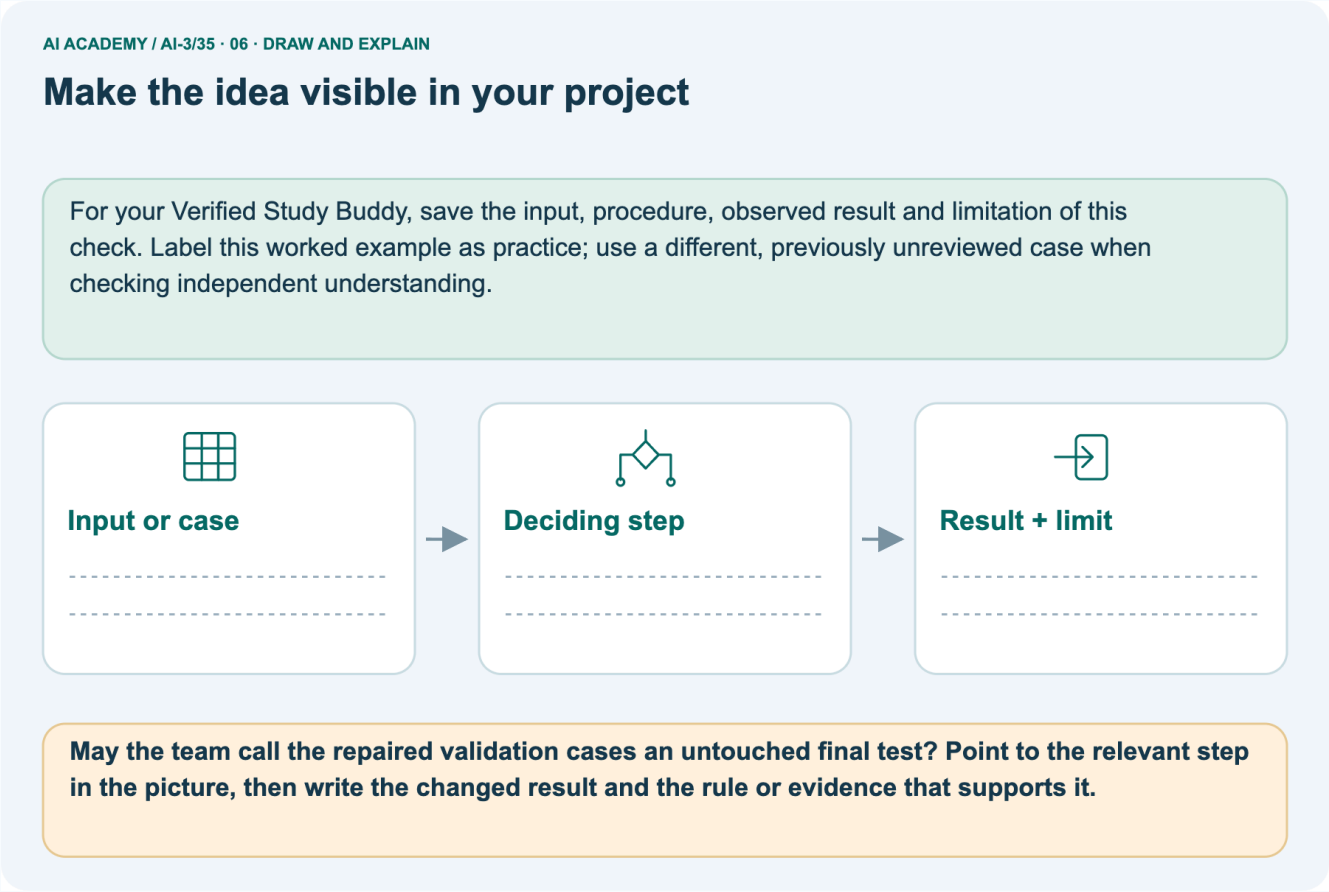
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Keep the stated scope: Passing a limited privacy test is bounded **evidence**, not a guarantee against all future disclosure.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: May the team call the repaired validation cases an untouched final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Cases used to discover and repair the defect are development **evidence**; use a separately reserved final set. Changed-case check: It passes this privacy gate and its validation task rate is 75%.

A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.